

helixqa-jackett-fake

helixqa_jackett_fake_behavioral_ equivalence_challenge.sh

Revision: 1 **Last modified:** 2026-08-15T12:00:00Z **Status:** active
Maintainer: Track 11 — BOB-067 Lava-P4 port **Scope:** anti-bluff
regression guard for the behaviorally-equivalent Jackett fake used by
the boba autonomous test regime

Overview

The Lava-P4 porting directive is explicit: *“the test fake MUST 302-
without- cookie like real Jackett (behavioral equivalence) or the gap stays
hidden.”* This challenge is the mechanized self-validation guard
proving the fake never degrades into a bluff-fake (a fake that would
200 on apikey-only management and hide the exact defect the fake was
built to catch).

Fake location

The fake is embedded in qBitTorrent-go/internal/jackett/
cookie_login_test.go as newPasswordProtectedJackett() (Go
httptest.Server). Staying in _test.go keeps it out of every production
import graph (§11.4.27 no-fakes- beyond-unit-tests) while remaining
consumable by every test in the same package.

Behavioral contract (what the fake MUST reproduce)

Cross-checked LIVE against real Jackett at <http://localhost:9117> on 2026-08-15 (values captured verbatim from `curl -s -D-`):

Behavior	Real Jackett (live)	Fake
GET /api/v2.0/indexers? apikey=<bogus> (no cookie)	HTTP/1.1 302 + Location: .../UI/ Login? ReturnUrl=%2Fapi%2Fv2.0%2Findexers...	HTTP 302 + Location: / UI/Login
POST /UI/ Dashboard password= (correct/empty)	HTTP/1.1 302 + Set-Cookie: Jackett=CfDJ8... + Location: Dashboard	HTTP 302 + Set-Cookie: Jackett=valid-session- token + Location: /
POST /UI/ Dashboard password=<wrong>	HTTP/1.1 200 re-renders login HTML, NO Set-Cookie	HTTP 200 + <html>login</ html>, NO Set-Cookie
GET /api/v2.0/ indexers WITH valid session cookie	HTTP/1.1 200 JSON catalog	HTTP 200 + fixture JSON [{"id":"rutracker",...}]

The cookie name Jackett matches real Jackett exactly (verified live).

Prerequisites

- go toolchain on PATH (challenge SKIPS with reason if absent).
- qBitTorrent-go/internal/jackett/cookie_login_test.go present.
- OPTIONAL: a live Jackett reachable at \$JACKETT_LIVE_URL (default <http://localhost:9117>). If reachable, the challenge diffs the fake's contract against the live product on B1 (apikey-only→302) and B2 (login→302+Set-Cookie: Jackett=). If not reachable, that cross-check is honestly SKIPPed with an explicit [UNCONFIRMED-AGAINST-LIVE-JACKETT] marker in the output.

Usage

```
# GREEN – shipped regression guard
RED_MODE=0 bash challenges/scripts/
             helixqa_jackett_fake_behavioral_equivalence_challenge.sh
```

```
# GREEN with an explicit live target
JACKETT_LIVE_URL=http://jackett.internal:9117 \
    bash challenges/scripts/
    helixqa_jackett_fake_behavioral_equivalence_challenge.sh

# RED – reproduces the bluff-fake / absent-fake state (PASS if
    degraded)
RED_MODE=1 bash challenges/scripts/
    helixqa_jackett_fake_behavioral_equivalence_challenge.sh
```

What it proves

1. Runs TestFakeJackettRefusesManagementWithoutCookie (already embedded next to the fake) — the load-bearing anti-bluff guard that a bare GET /api/v2.0/indexers?apikey=... (no cookie) returns 302, not 200.
2. Asserts the pass line is present with -v output (§11.4.201 false-null guard: a mis-typed -run filter also returns exit 0 with no test).
3. When a live Jackett is reachable, cross-checks the fake's B1 + B2 against the real product using raw curl (independent transport, no shared code path with the fake or the client-under-test).

Falsifiability rehearsal (paired §1.1 mutation)

The sibling jackett_cookie_login_hardening_challenge.sh doc describes a doManaged-side mutation that made the challenge FAIL. Under this challenge, the analogous fake-side mutation is: remove the hasValidSession(r) gate from the management branch so the fake returns 200 on apikey-only management. TestFakeJackettRefusesManagementWithoutCookie catches that mutation directly and this challenge exits 1.

Live evidence (2026-08-15, localhost:9117)

```
$ curl -s -D- -o /dev/null --max-time 3 \
    'http://localhost:9117/api/v2.0/indexers?apikey=bogus-
    key&configured=false'
HTTP/1.1 302 Found
Location: http://localhost:9117/UI/Login?
ReturnUrl=%2Fapi%2Fv2.0%2Findexers%3Fapikey%3Dbogus-
key%26configured%3Dfalse
```

```
$ curl -s -D- -o /dev/null --max-time 3 -X POST -d 'password=' \
  http://localhost:9117/UI/Dashboard
HTTP/1.1 302 Found
Location: Dashboard
Set-Cookie: Jackett=CfDJ8J4t0dXrfJhAr2dBxs4c...; path=/;
samesite=lax; httponly
```

The fake's Location/Set-Cookie shape matches — behavioral equivalence is PROVEN LIVE, not asserted.

Honest boundary (§11.4.6)

- Live-check verifies B1 + B2. B3 (wrong-password → 200 no-cookie) is covered by the Go test `TestManagementCookieLogin_ConfigurableAdminPassword` rather than a live probe (a live probe with a wrong password would either lock the account or be blocked as brute-force). B4 (valid-cookie → 200 JSON) is covered by the Go tests that consume the fake.
- The fake reproduces the discriminating BEHAVIOR (cookie gate, status codes, Set-Cookie name); it does NOT reproduce Jackett's full API surface — endpoints beyond `/UI/Dashboard`, `/UI/Login`, `/api/v2.0/indexers`, `/api/v2.0/indexers/*/config` return 404. Extending the fake is a follow-up when a new consumer is added.

Cross-references

- Sibling: `docs/scripts/jackett-cookie-login-hardening.md`.
- Fake source: `qBitTorrent-go/internal/jackett/cookie_login_test.go`.
- Client-under-test: `qBitTorrent-go/internal/jackett/client.go`.
- Port doc: `docs/PORTING-FROM-LAVA.md (P4)`.
- Constitution: §11.4.107(10) / §11.4.115 / §11.4.201 / §11.4.238.